



Clouds — FULL MIND MAP

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Chapter 40_Clouds

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Clouds — Formation aur Types

- **Condensation** se clouds banti hain — water gaseous se liquid/solid state mein convert hoti hai, atmosphere mein tiny water droplets/ice crystals ka collection cloud kehlata hai. [Q1]
- **Cumulonimbus** clouds highly intense rain ke liye responsible hain — highly developed+dense, rain/hail/thunderstorm ki high possibility, ~1 hour mein dissipate ho jaati hain (short-lived heavy rain). [Q2]
- Given options mein highest altitude cloud: **Cirrostratus** (upper troposphere mein) — **Altostratus**, **Altostratus**, **Cumulus** isse neeche hoti hain. [Q3]

Thunder Clouds — Development Factors

- Thunder clouds development factors: **Condensation, High temperature+humidity, Vertical wind** — **Orography factor NAHI** hai. High temp se strong upward winds, jo water droplets ko upar le jaati hain (freeze/condense), phir wapas girti hain — lightning+sound generate hota hai. [Q4]

Thunder — Sound ka Actual Cause

- Thunder (garjan) actually **lightning discharge se hoti hai** — lightning air ko heat karti hai, sudden overpressure banti hai, shock wave supersonically expand hoti hai — cumulonimbus clouds ka meeting, nimbus clouds separate hona, ya sirf air/water particles ki upward movement — in teeno options mein se KOI bhi sahi explanation NAHI hai. [Q5]

Cirrus Cloud Thinning — Climate Engineering

- Cirrus cloud thinning technique aur stratosphere mein sulphate aerosol injection ka suggestion **Global Warming kam karne** ke context mein hai — cirrus clouds ice crystals se bane hote hain, ground se reflected/emitted heat ko space mein escape hone se rokte hain; thinning se heat-absorption capacity kam hoti hai. Sulphate aerosol solar radiation ka part reflect karta hai troposphere tak pahunchne se pehle. [Q6]

Cloudy Night — Dewdrops Formation

- Cloudy night pe dew NAHI banti kyunki **clouds Earth ki outgoing infrared radiation ko absorb karke wapas reflect/re-emit kar dete hain** — surface strongly cool nahi hota, dew-point tak nahi pahunchta. [Q7]